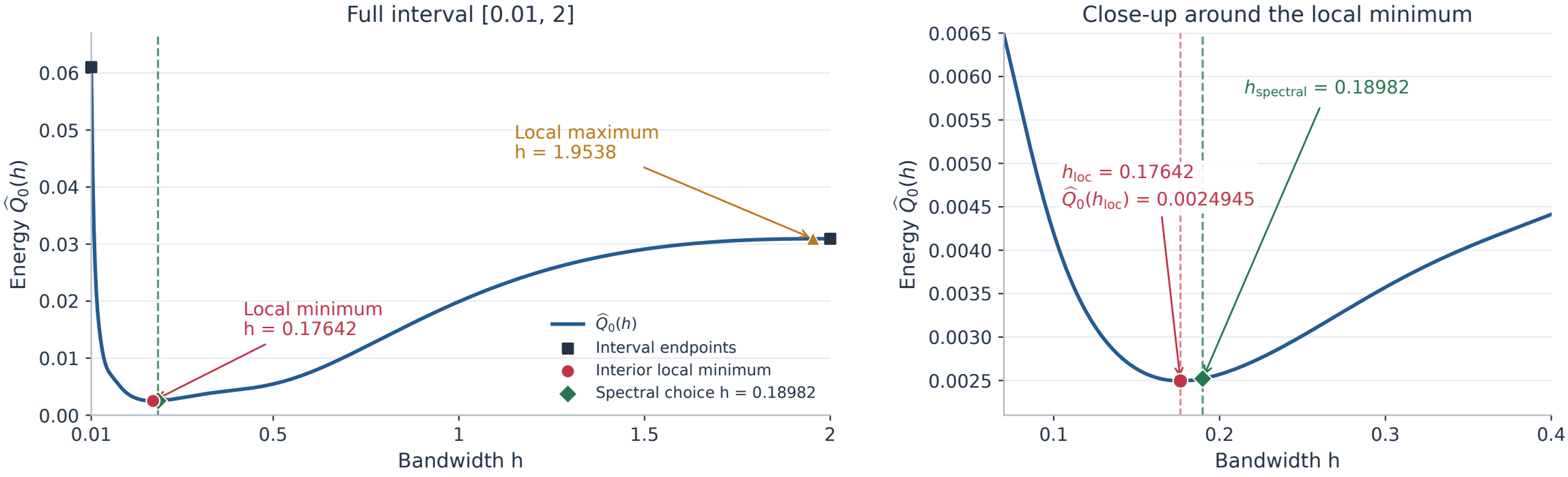
$$\widehat{Q}_0(h) = \frac{h^2}{2}\,\|\partial_h \hat{f}_h\|_2^2$$

Gaussian kernel KDE | fixed reference sample, n = 200 | N = 100, geometric grid, $\lambda_N = N^{3/2} = 1000$



The marked local minimum is the lowest detected energy on [0.01, 2], including both endpoints.

Complete energy search: 102.6 ms (median of 9 runs after initialization; includes refinement and endpoint comparison).

Spectral selection: 12.6 ms (median of 9 runs; includes Fourier matrix construction and eigensolve; one numerical thread).